Identities used: binomial therom
$(x+a)^r = \sum_{b=0}^{r} \left[\binom{r}{b} x^b a^{r-b} \right]$
pascals identity
$\binom{a}{n} = \binom{a-1}{n} + \binom{a-1}{n-1}$

Reucrsions from Binomial therom

Non falling sums

$$(x+a)^r = \sum_{b=0}^{r} \left[\binom{r}{b} x^b a^{r-b} \right]$$

let $a = 1$

$$(x+1)^r = \sum_{b=0}^{r} \left[\binom{r}{b} x^b \right]$$

let $x = x' - 1$

$$((x'-1)+1)^r = x^r = \sum_{b=0}^{r} \left[\binom{r}{b} (x'-1)^b \right]$$

in general

$$((x-b)+a)^r = \sum_{b=0}^{r} \left[\binom{r}{b} (x-b)^b a^{r-b} \right]$$

if b=a

$$x^r = \sum_{b=0}^{r} \left[\binom{r}{b} (x-a)^b a^{r-b} \right]$$

we can subtitute the defintion of x^r into $(x-a)^b$ or a^{r-b}
$(x-a)^b$ is a good substitution as it the nested power term will shrink by a
after $\frac{x}{a}$ expansions the inner power term will be 0

$$x^r = \sum_{b=0}^{r} \left[\binom{r}{b} (x-a)^b a^{r-b} \right]$$

$$(x-a)^b = \sum_{c=0}^{b} \left[\binom{b}{c} ((x-a)-a)^c a^{b-c} \right]$$

$$x^r = \sum_{b=0}^{r} \left[\binom{r}{b} \sum_{c=0}^{b} \left[\binom{b}{c} ((x-a)-a)^c a^{b-c} \right] a^{r-b} \right]$$

every layer strips one a from inner power's base

$$x^r = \sum_{b=0}^{r} \left[\binom{r}{b} \sum_{c=0}^{b} \left[\binom{b}{c} ((x-a)-a)^c a^{b-c} \right] a^{r-b} \right]$$

$$x^r = \overbrace{\sum_{b=0}^{r} \left[\binom{r}{b} \sum_{c=0}^{b} \left[\binom{b}{c} \sum_{d=0}^{c} \left[\binom{c}{d} \dots \left[\sum_{n_{\frac{x}{a}}=0}^{n_{(\frac{x}{a})-1}} \left[\binom{n_{(\frac{x}{a})-1}}{n_{\frac{x}{a}}} 0^{n_{\frac{x}{a}}} a^{n_{(\frac{x}{a})-1}-n_{\frac{x}{a}}} \right] \right] a^{c-d} \right] a^{b-c} \right] a^{r-b} \right]}^{\frac{x}{a} \text{ sums}}$$

you can stop at any layer you want, you dont need to fully expand

$$x^r = \overbrace{\sum_{b=0}^{r} \left[\binom{r}{b} \sum_{c=0}^{b} \left[\binom{b}{c} \sum_{d=0}^{c} \left[\binom{c}{d} \dots \left[\sum_{n_l=0}^{n_{l-1}} \left[\binom{n_{l-1}}{n_l} (x-la)^{n_l} a^{n_{l-1}-n_l} \right] \right] a^{c-d} \right] a^{b-c} \right] a^{r-b} \right]}^{l \text{ sums}}$$

falling sums

$$(x+a)^r = \sum_{b=0}^{r} \left[\binom{r}{b} x^b a^{r-b} \right]$$

$$(x+a)^r = \sum_{b=0}^{r-1} \left[\binom{r}{b} x^b a^{r-b} \right] + \binom{r}{r} x^r a^0$$

$$((x-a)+a)^r = x^r = \sum_{b=0}^{r-1} \left[\binom{r}{b} (x-a)^b a^{r-b} \right] + \binom{r}{r} (x-a)^r a^0$$

the same reassoning applys here as in the last case:
"we can subtitute the defintion of x^r into $(x-a)^b$ or a^{r-b}
$(x-a)^b$ is a good substitution as it the nested power term will shrink by a
after $\frac{x}{a}$ expansions the inner power term will be 0 "

$$x^r = \sum_{b=0}^{r-1}\left[\binom{r}{b}a^{r-b}(x-a)^b\right] + \binom{r}{r}a^0(x-a)^r$$

$$x^r = \sum_{b=0}^{r-1}\left[\binom{r}{b}a^{r-b}\sum_{c=0}^{b-1}\left[\binom{b}{c}a^{b-c}((x-a)-a)^c\right] + \binom{b}{b}a^0((x-a)-a)^b\right] + \binom{r}{r}(x-a)^r a^0$$

$$x^r = \overbrace{\sum_{b=0}^{r-1}\left[\binom{r}{b}a^{r-b}\sum_{c=0}^{b-1}\left[\binom{b}{c}a^{b-c}\right.\right.}^{l \text{ sums}}$$
$$\left.\left.\cdots\left[\sum_{n_l=0}^{n_{l-1}-1}\left[\binom{n_{l-1}}{n_l}a^{n_{l-1}-n_l}(x-la)^{n_l}\right] + \binom{n_{l-1}}{n_{l-1}}a^0(x-la)^n_{l-1}\right]\cdots\right.\right.$$
$$\left.\left. + \binom{b}{b}a^0((x-a)-a)^b\right]\right] + \binom{r}{r}(x-a)^r a^0$$

$$a^0 = 1 \text{ for all } a \in \mathbb{R}$$

$$\binom{r}{r} = 1 \text{ for all } r \in \mathbb{Z}^+$$

if we restrict our formula with these we can simplify it

according to our restriciton $a^0 = 1$ and $\binom{r}{r} = 1$

$$x^r = \overbrace{\sum_{b=0}^{r-1}\left[\binom{r}{b}a^{r-b}\left(\sum_{c=0}^{b-1}\left[\binom{b}{c}a^{b-c}\left(\cdots\sum_{n_l=0}^{n_{l-1}-1}\left[\binom{n_{l-1}}{n_l}a^{n_{l-1}-n_l}(x-la)^{n_l}\right] + (x-la)^{n_{l-1}}\cdots\right)\right] + (x-2a)^b\right)\right]}^{l \text{ sums}} + (x-a)^r$$

the sum can be factored into a sum of falling sums The papers theory holds if the restrictions hold:

$$a^0 = 1 \text{ for all } a \in \mathbb{R}$$

$\binom{r}{r} = 1$ for all $r \in \mathbb{Z}^+$ Since we can freely choose a only the restriction on r remains

“I have a theory, all x^n’s can be written as a sum of nested sums with falling upper bounds “

all x^r’s can be written as a sum of nested sums with falling upper bounds for all $r \in \mathbb{Z}^+$ also x must have a divisor d in $\mathbb{R}$ where $\left(\frac{x}{d} = l, l \in \mathbb{Z}^+\right)$, if it has this divisor it can be written a nested sum with l sums for all $a \in \mathbb{R} \mid r \in \mathbb{Z}^+ \mid d$ in $\mathbb{R}$ where $\left(\frac{x}{d} = l, l \in \mathbb{Z}^+\right) \Rightarrow$

$$x^r = \overbrace{\sum_{b=0}^{r-1}\left[\binom{r}{b}a^{r-b}\left[\sum_{c=0}^{b-1}\left[\binom{b}{c}a^{b-c}\cdots\left[\sum_{n_l=0}^{n_{l-1}-1}\left[\binom{n_{l-1}}{n_l}a^{n_{l-1}-n_l}(x-la)^{n_l}\right]\right]\cdots\right]\right]\right]}^{l \text{ sums}}$$
$$+\overbrace{\sum_{b=0}^{r-1}\left[\binom{r}{b}a^{r-b}\left[\sum_{c=0}^{b-1}\left[\binom{b}{c}a^{b-c}\cdots\left[\sum_{n_{l-1}=0}^{n_{l-2}-1}\left[\binom{n_{l-2}}{n_{l-1}}a^{n_{l-2}-n_{l-1}}(x-la)^{n_{l-1}}\right]\right]\cdots\right]\right]\right]}^{l-1 \text{ sums}}$$
$$+\cdots+\overbrace{\sum_{b=0}^{r-1}\left[\binom{r}{b}a^{r-b}\left[(x-2a)^b\right]\right]}^{1 \text{ sums}}$$
$$+\overbrace{(x-a)^r}^{0 \text{ sums}}$$

Falling sums (l,r upper bound)

$$(x+a)^r = \sum_{b=0}^{r}\left[\binom{r}{b}x^b a^{r-b}\right]$$

$$(x+a)^r = \sum_{b=0}^{r-1}\left[\binom{r}{b}x^b a^{r-b}\right] + \binom{r}{r}x^r a^0$$

$$\text{let } x' = x-a, \text{let } a' = a$$

$$((x-a)+a)^r = \sum_{b=0}^{r-1}\left[\binom{r}{b}(x-a)^b a^{r-b}\right] + \binom{r}{r}(x-a)^r a^0$$

the previous falling sums resulted from expanding the inner (x-a)^b, if we expand the outer (x-a)^r we get a diffrent falling sum

$$((x-a)+a)^r = x^r = \sum_{b=0}^{r-1}\left[\binom{r}{b}(x-a)^b a^{r-b}\right] + \binom{r}{r}(x-a)^r a^0$$

$$x^r = \sum_{b=0}^{r-1}\left[\binom{r}{b}(x-a)^b a^{r-b}\right] + \binom{r}{r}\left[\sum_{b=0}^{r-1}\left[\binom{r}{b}((x-a)-a)^b a^{r-b}\right] + \binom{r}{r}((x-a)-a)^r a^0\right]a^0$$

$$x^r = \overbrace{\sum_{b=0}^{r-1}\left[\binom{r}{b}(x-a)^b a^{r-b}\right] + \binom{r}{r}\left[\sum_{b=0}^{r-1}\left[\binom{r}{b}((x-a)-a)^b a^{r-b}\right] + \binom{r}{r}\right.}^{l \text{ sums}}$$

$$\left.\cdots\left[\sum_{b=0}^{r-1}\left[\binom{r}{b}(x-la)^b a^{r-b}\right] + \binom{r}{r}(x-la)^r a^0\right]\cdots a^0\right]a^0$$

$$x^r = \overbrace{\sum_{b=0}^{r-1}\left[\binom{r}{b}(x-a)^b a^{r-b}\right]}^{l \text{ sums}}$$

$$+\binom{r}{r}a^0\sum_{b=0}^{r-1}\left[\binom{r}{b}(x-2a)^b a^{r-b}\right]$$

$$+\binom{r}{r}^2 a^{(0)^2}\sum_{b=0}^{r-1}\left[\binom{r}{b}(x-3a)^b a^{r-b}\right]$$

$$+\cdots+\binom{r}{r}^{l-1} a^{(0)^{l-1}}\sum_{b=0}^{r-1}\left[\binom{r}{b}(x-la)^b a^{r-b}\right]$$

$$+\binom{r}{r}^{l}(x-la)^r a^{(0)^l}$$

$$x^r = \sum_{b=0}^{l-1}\left[\binom{r}{r}^b a^{(0)^b}\sum_{c=0}^{r-1}\left[\binom{r}{c}(x-(b+1)a)^c a^{r-c}\right]\right] + \binom{r}{r}^l(x-la)^r a^{(0)^l}$$

note: for the simplifed sum aboe to work binom(r,r)^(0) and a^(0)^(0) must simplify to 1, so a^0 is implictly assumed to be 1 for that sum to work. you could factor out first term of the sum to make it pure

if we add restrictions the sum simplifies

$$a^0 = 1 \text{ for all } a \in \mathbb{R}$$

$$\binom{r}{r} = 1 \text{ for all } r \in \mathbb{Z}^+$$

for all $a \in \mathbb{R} \mid r \in \mathbb{Z}^+ \mid l \in \mathbb{Z}^+$

$$x^r = \sum_{b=0}^{l-1} \left[\sum_{c=0}^{r-1} \left[\binom{r}{c} (x-(b+1)a)^c a^{r-c} \right] \right] + (x-la)^r$$

Falling sums (l,r upper bound) reversed

we can reverse the sum
(x-(b+1)a) is isolated from the inner sum
if b starts at l-1 and goes upto x-1a the sum is still identical but reversed

for all $a \in \mathbb{R} \mid r \in \mathbb{Z}^+ \mid l \in \mathbb{Z}^+$

$$x^r = \sum_{b=(l-1)}^{x-a} \left[\sum_{c=0}^{r-1} \left[\binom{r}{c} (b)^c a^{r-c} \right] \right] + (x-la)^r$$

falling sums, free upper bounds

shrinking terms

if we just keep exapnding the new outer term derived from the falling l,r formula with the falling l,r formula we get more nested sums

for all $a \in \mathbb{R} \mid r \in \mathbb{Z}^+ \mid l \in \mathbb{Z}^+$

$$x^r = \sum_{b=0}^{l-1} \left[\sum_{c=0}^{r-1} \left[\binom{r}{c} (x-(b+1)a)^c a^{r-c} \right] \right] + (x-la)^r$$

$$x^r = \sum_{b=0}^{l-1} \left[\sum_{c=0}^{r-1} \left[\binom{r}{c} (x-(b+1)a)^c a^{r-c} \right] \right] + \sum_{b=0}^{l-1} \left[\sum_{c=0}^{r-1} \left[\binom{r}{c} ((x-la)-(b+1)a)^c a^{r-c} \right] \right] + ((x-la)-la)^r$$

$$x^r = \sum_{d=0}^{m-1} \left[\sum_{b=0}^{l-1} \left[\sum_{c=0}^{r-1} \left[\binom{r}{c} ((x-dla)-(b+1)a)^c a^{r-c} \right] \right] \right] + (x-mla)^r$$

$$x^r = \sum_{b_3=0}^{n_3-1} \left[\sum_{b_2=0}^{n_2-1} \left[\sum_{b_1=0}^{n_1-1} \left[\sum_{b_0=0}^{r-1} \left[\binom{r}{b_0} (((x-b_3n_2n_1n_0)-b_2n_1n_0)-(b_1+1)n_0)^{b_0} n_0^{r-b_0} \right] \right] \right] \right] + (x-n_3n_2n_1n_0)^r$$

if we expand n times

$$x^r = \sum_{b_n=0}^{n_n-1} \left[\dots \sum_{b_4=0}^{n_4-1} \left[\sum_{b_3=0}^{n_3-1} \left[\sum_{b_2=0}^{n_2-1} \left[\sum_{b_1=0}^{n_1-1} \left[\sum_{b_0=0}^{r-1} \left[\right. \right. \right. \right. \right. \right.$$

$$\binom{r}{b_0} ((((\dots(x-b_nn_{n-1}\dots n_3n_2n_1n_0)\dots - b_4n_3n_2n_1n_0) - b_3n_2n_1n_0) - b_2n_1n_0) - (b_1+1)n_0)^{b_0} n_0^{r-b_0}$$

$$\left. \left. \left. \left. \left. \left. \right] \right] \right] \right] \right] \right] + (x-n_n\dots n_4n_3n_2n_1n_0)^r$$

for all $r \in \mathbb{Z}^+ \mid n_0 \in R \mid n_0 \neq 0 \mid n_1 \dots n_n \in \mathbb{Z}^+$

$$x^r = \overbrace{\sum_{b_n=0}^{n_n-1} \left[\dots \sum_{b_2=0}^{n_2-1} \left[\sum_{b_1=0}^{n_1-1} \left[\sum_{b_0=0}^{r-1} \left[\binom{r}{b_0} ((\dots(x-b_nn_{n-1}\dots n_2n_1n_0)\dots - b_2n_1n_0) - (b_1+1)n_0)^{b_0} n_0^{r-b_0} \right] \right] \right] \right]}^{n \text{ sums}} + (x-n_n\dots n_2n_1n_0)^r$$

growing terms

this can be cleaned up if we avoid $((x-a)+a)^r = x^r$ substituion

$$(x+a)^r = \sum_{b_0=0}^{r-1}\left[\binom{r}{b_0}x^{b_0}a^{r-b_0}\right] + \binom{r}{r}a^0x^r$$

$$x^r = -\sum_{b_0=0}^{r-1}\left[\binom{r}{b_0}x^{b_0}a^{r-b_0}\right] + (x+a)^r$$

$$x^r = -\sum_{b_0=0}^{r-1}\left[\binom{r}{b_0}x^{b_0}a^{r-b_0}\right] + \left[-\sum_{b_0=0}^{r-1}\left[(x+a)^{b_0}a^{r-b_0}\right] + ((x+a)+a)^r\right]$$

$$x^r = \sum_{b_1=0}^{n_1-1}\left[-\sum_{b_0=0}^{r-1}\left[\binom{r}{b_0}(x+b_1a)^{b_0}a^{r-b_0}\right]\right] + ((x+n_1a)^r$$

$$x^r = \sum_{b_1=0}^{n_1-1}\left[-\sum_{b_0=0}^{r-1}\left[\binom{r}{b_0}(x+b_1n_0)^{b_0}n_0^{r-b_0}\right]\right] + ((x+n_1n_0)^r$$

the restrictions we have are from $\binom{r}{r}(n_0)^0$ being applied to each new varible. factoring out the -1 we get:

$$\text{for all } r \in \mathbb{Z}^+ \mid n_0 \in R \mid n_0 \neq 0 \mid n_1 ... n_n \in \mathbb{Z}^+$$

$$x^r = -1\overbrace{\sum_{b_n=0}^{n_n-1}\left[...\sum_{b_2=0}^{n_2-1}\left[\sum_{b_1=0}^{n_1-1}\left[\sum_{b_0=0}^{r-1}\left[\binom{r}{b_0}(((x+b_nn_n...n_1n_0)+...+b_2n_1n_0)+b_1n_0)^{b_0}n_0^{r-b_0}\right]\right]\right]\right]}^{n \text{ sums}} + ((x+n_n...n_2n_1n_0)^r$$

binomial recurssion

hockey stick

pascals identity: $\binom{a}{n} = \binom{a-1}{n} + \binom{a-1}{n-1}$

pasclas identity can be applied to its own terms

i will apply the idenitty to its left hand terms

$$\binom{a}{n} = \binom{a-1}{n} + \binom{a-1}{n-1}$$

$$\binom{a}{n} = \left[\binom{a-2}{n} + \binom{a-2}{n-1}\right] + \binom{a-1}{n-1}$$

$$\binom{a}{n} = \left[\left[...\left[\binom{a-a}{n} + \binom{a-a}{n-1}\right]... + \binom{a-3}{n-1}\right] + \binom{a-2}{n-1}\right] + \binom{a-1}{n-1}$$

$$\binom{a}{n} = \sum_{b=0}^{a-1}\left[\binom{b}{n-1}\right] + \binom{0}{n}$$

the hockey stick identity: binom_hockey

hockey stick recurssion

the hockey stick identity can also be applied to its own terms o get a nested sum with falling uper bounds

$$\binom{a}{n} = \sum_{b=0}^{a-1}\left[\binom{b}{n-1}\right] + \binom{0}{n}$$

$$\binom{a}{n} = \sum_{b=0}^{a-1}\left[\sum_{c=0}^{b-1}\left[\binom{c}{n-2}\right] + \binom{0}{n-1}\right] + \binom{0}{n}$$

$$\binom{a}{n} = \sum_{b=0}^{a-1}\left[\sum_{c=0}^{b-1}\left[\cdots \sum_{n_n=0}^{n_{n-1}-1}\left[\binom{n_n}{n-n}\right] + \binom{0}{n-(n-1)}\cdots\right] + \binom{0}{n-1}\right] + \binom{0}{n}$$

if we restrict n the sum simplifies

this simplifes the $\binom{0}{n-d}$ at each layer

for all $n \in \mathbb{Z}_{\geq 0}$

$$\binom{0}{n} = \begin{cases} 1 \text{ if } n = 0 \\ 0 \text{ if } n \neq 0 \end{cases}$$

this simplifes the innermost $\binom{n_n}{0}$

for all $a \in \mathbb{R}$

$$\binom{a}{0} = 1$$

when $n \neq 0$ the inner $\binom{0}{n-d}$ simplifies to 0, when n=0, the inner sums are never reached and $\binom{a}{0} = \binom{0}{0}$ hockey stick recurssion identity:

for all $n \in \mathbb{Z}_{\geq} 0$

$$\binom{a}{n} = \overbrace{\sum_{b=0}^{a-1}\left[\sum_{c=0}^{b-1}\left[\cdots \sum_{n_n=0}^{n_{n-1}-1}[1]\cdots\right]\right]}^{n \text{ sums}} + \binom{0}{n}$$